

Transformer-based Language Models



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- **Encoder-only Transformer Models**

- **Masked Language Models:** Designed to consider bidirectional contexts that are beneficial to classification and language understanding
- **BERT Series:** BERT (2018-10), DistilBERT, RoBERTa, XLM, XLM-RoBERTa, ALBERT, ELECTRA

- **Decoder-only Transformer Models**

- **Autoregressive Language Models:** Designed to predict the next token (sub-word), suitable for text generation
- **GPT Series:** GPT-1 (2018-6), GPT-2 (2019-02), GPT-3 (2020-06), InstructGPT (2022-02), ChatGPT (2022-11), GPT-4 (2023-03), GPT-4 Turbo (2023-11)

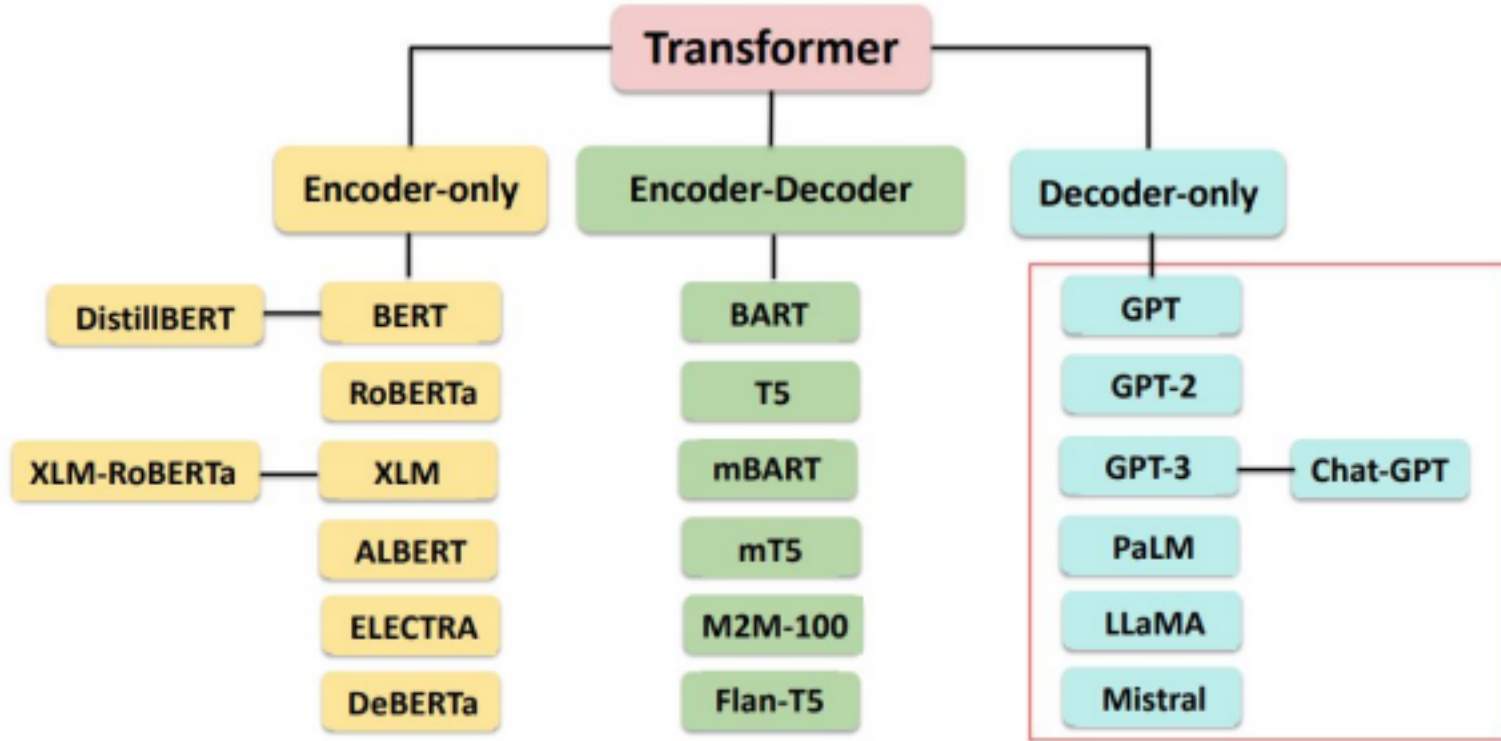
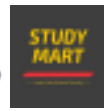
- **Encoder-Decoder Transformer Models**

- **Sequence-to-sequence models**
- **Original Transformer** (2017-06), **T5** (2019-10), **BART** (2019-11), **M2M-100** (2021-07), **Flan-T5** (2022-10)

Transformer-based Models

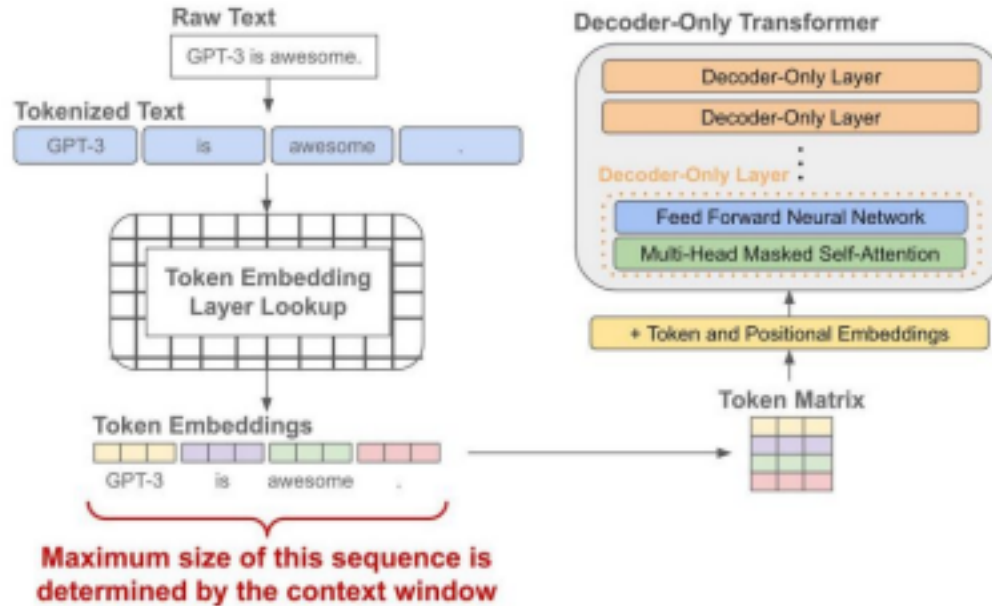


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Decoder-Only

Transformer



Notably, the decoder-only architecture **removes not only the encoder, but also all of the cross-attention layers in the decoder**, as there is no longer any encoder to which the cross-attention layers can attend.

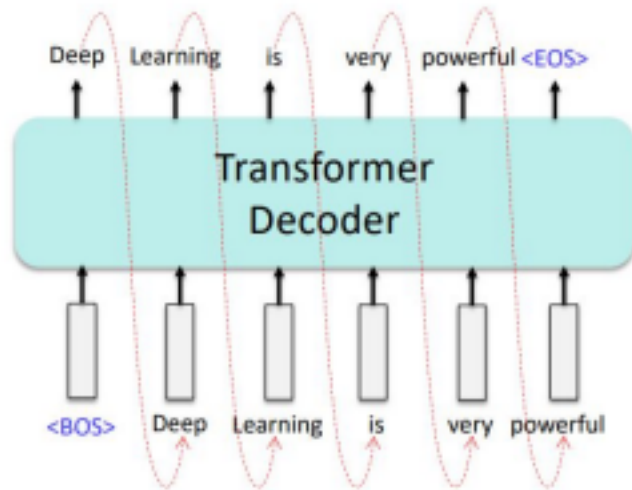
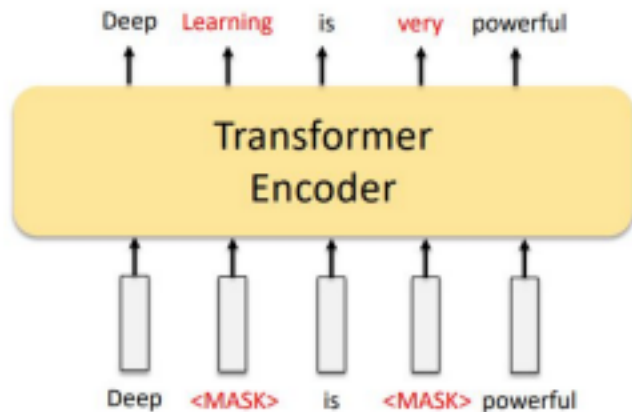
Why Decoder-only Transformer Models?



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- **Encoder-only** Transformer models like BERT are **NOT suitable for Text Generation applications**. This is because they do not naturally lead to generation methods.
- In contrast, **Decoder-only** Transformer models are naturally **autoregressive model** (aka left-to-right model), which are **well-suited for Text Generation Tasks**.



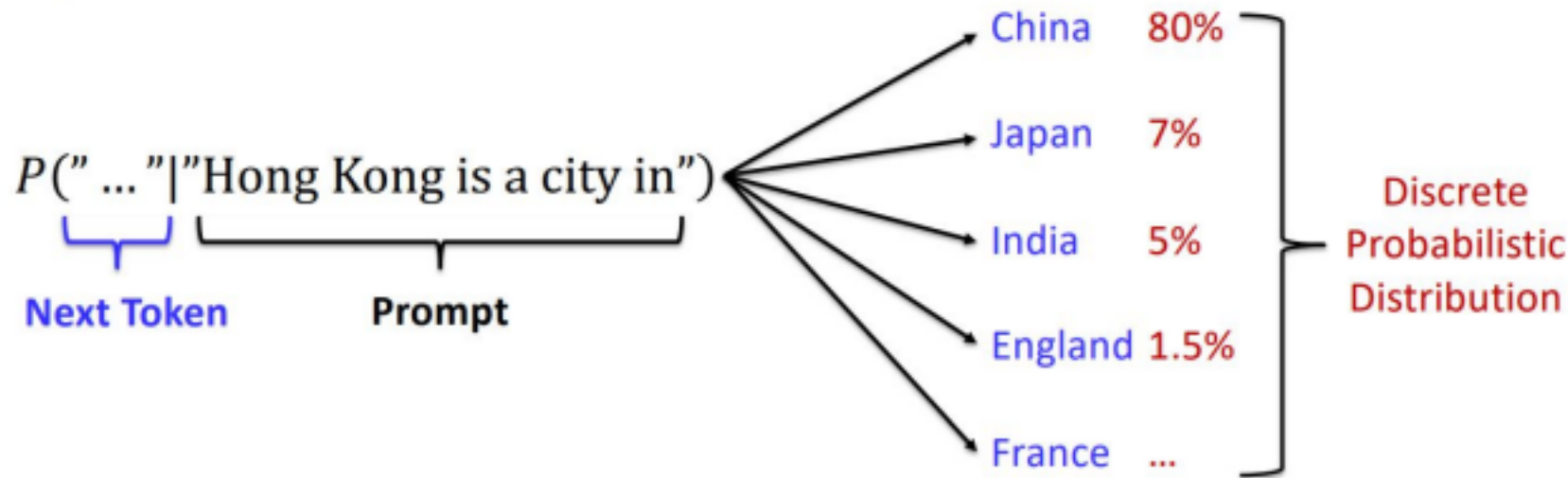
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- It is common to refer to **Language Models** as "**Autoregressive Language Models**," which are **Probabilistic Models** that assign probabilities to sequences of words.



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Autoregressive

Language

Models





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Autoregressive Language Models



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Generative Pre-Trained Transformer (GPT)





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Evolution of OpenAI GPT Models





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GPT (2018-06)





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GPT

Architecture

and

Pretraining





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Supervised Fine Tuning of GPT





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GPT-2 (2019-02) - Introduced Zero-Shot Prompting





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GTP-2:

Emergent

Zero-Shot

Learning





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GTP-2:

Emergent

Zero-Shot

Learning





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GTP-2: Emergent Zero-Shot Learning





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GPT-2

Implementation

Example





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GPT-3 (2020-06) - Few Shot Prompting / In Context Learning





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GPT-3: Emergent Few-Shot Learning





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GPT-3: In-Context Learning





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GPT3: Fine-Tuning vs In-Context Learning





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GPT-3 Performance in SuperGLUE

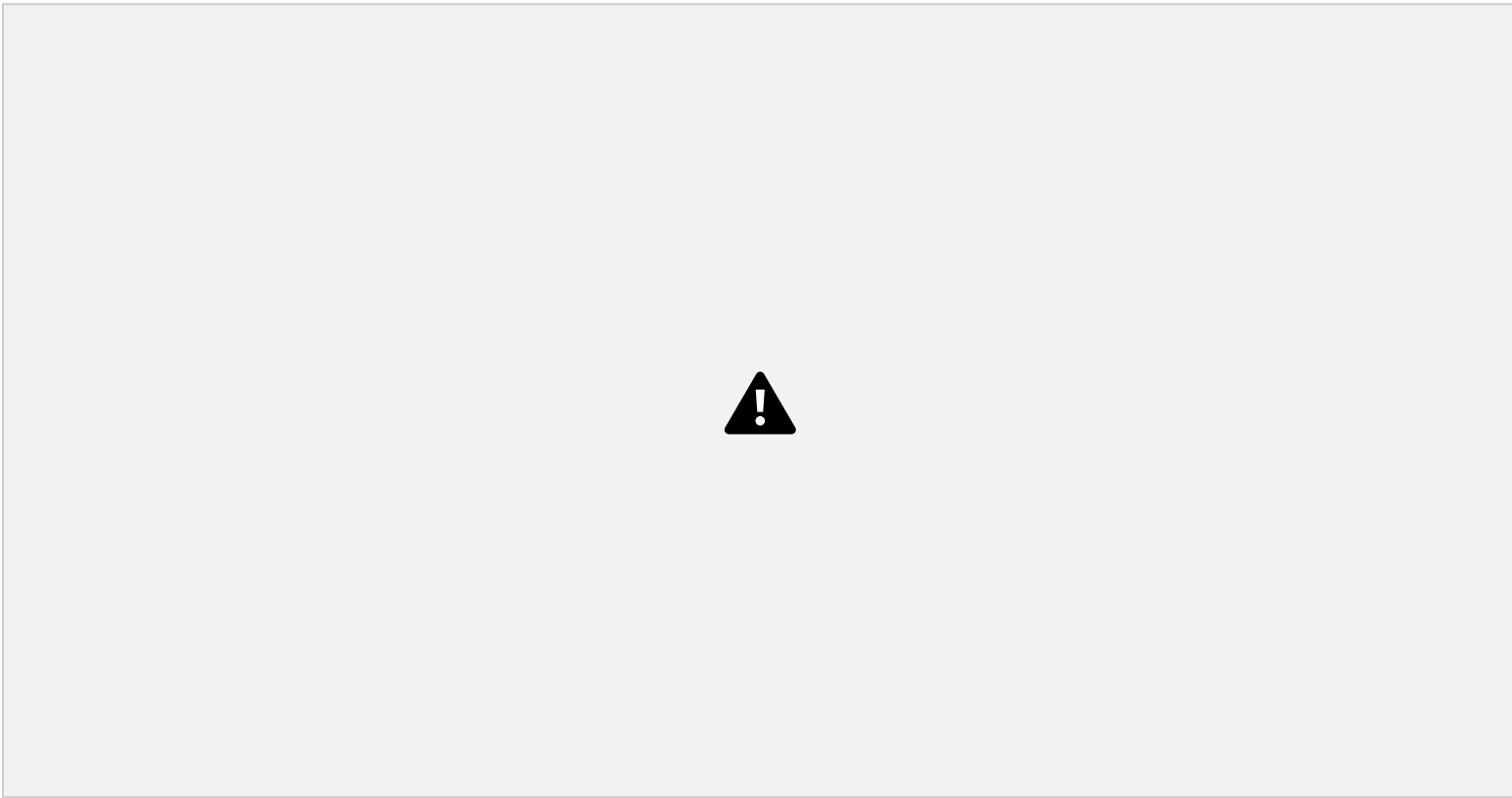
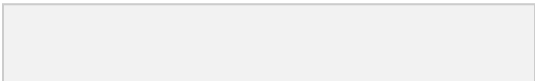




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Increasing the shot count increase performance in GPT3





Few-Shot Learning is an Emergent Property of Mode scale



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Pretraining: GPT vs BERT





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GPT-3 (Prompting) vs BERT (Pretrain-Finetune)





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Instruction Tuning and Model Alignment Trainings





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OpenAI's GPT LLM Series





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Claude LLM Series





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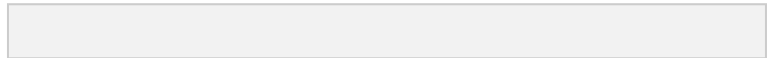
LLaMA LLM Series





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LLaMA LLM Series





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LLaMA LLM Series - LayerNorm vs RMSNorm





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LLaMA LLM Series - Grouped-Query Attention



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LLaMA LLM Series - Grouped-Query Attention





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LLaMA Family



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Mistral LLM Series





Mistral LLM Series - Mixtral-8x7B: Mixture-of-Experts (MoE)





Google's LLM Series



Google's LLM Series - Slide-Window Attention: Gemma 2



Baidu's ERNIE (文心一言) LLM Series



Alibaba's

Qwen

(通义千问)

LLM

Series



Huawei's

Pangu

(盘古)

LLM

Series



Closed-Source, Open-Source, Open-Weight LLMs



Closed-Source,

Open-Source,

Open-Weight

LLMs



Hugging Face LLM Performance Leaderboard



<https://huggingface.co/spaces/ArtificialAnalysis/LLM-Performance-Leaderboard>

Designing LLM Prompts for Different Tasks



Prompts - Text Summarization



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Prompts - Question Answering





Prompts - Text Classification





Prompts - Role Playing





Code

Generation



Reasoning



Prompt Engineering Techniques



Few-shot Prompts





Chain-of-Thought (CoT) Prompting



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Zero-Shot CoT





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Self-Consistency



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Self-Consistency Example



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Self-Consistency Example





Generate Knowledge Prompting



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Generate Knowledge Prompting Example



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Generate Knowledge Prompting Example



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Program-aided Language Model (PAL)







ReAct



ReAct



Directional Stimulus Prompting



Directional Stimulus Prompting



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Directional Stimulus Prompting



